Response to Stakeholder Thread: Eluno POS v1

To: Rohan (Founder), Meena (Retail Head), Dr. Iyer (Optometry Manager), Vikram (Finance), Farhan (Ops/Fulfilment)
From: Priya (Product) & POS Engineering Team | **Date:** 4 Nov 2026 | **Subject:** Scoped Plan & Clarifications for POS v1 (Go-Live: 15 Nov)

Team, to replace Shopify POS across all 7 stores by 15 November, our priority is **uncompromising correctness**: every rupee, tax paisa, and inventory movement must reconcile with Shopify and tax authorities. Speed will be achieved through clean architecture without bypassing audit controls.

1. Key Questions for Stakeholders

Stakeholder	Question	Why It Matters for v1
Vikram (Finance)	Confirm exact FY naming format (`ELU/{STORE}/FY26/{NO}`), GST e-invoicing (IRN) / B2B/B2C rules, and whether cancelled invoices need tax-portal filing or internal ledger retention.	Dictates local sequence schemas and statutory audit compliance.
Retail & Finance	When discount rules conflict (e.g. higher-priority non-stackable offer vs. stackable set yielding greater savings), does the till maximize customer benefit or follow strict priority?	Dictates the optimization logic of the automated offer engine.
Vikram & Meena	Meena requested up to 10% manager discount; Vikram mandated reason codes and approver identity. Can Retail and Finance confirm: (a) 10% cap approval, (b) reason codes, and (c) authentication via `staff_id` with `can_override_discount: true` in `staff.json`?	Balances floor agility with audit compliance without shared PINs.
Dr. Iyer (Optometry)	If a customer's prescription is missing or expired (`valid_until` passed), should the till hard-block lens configuration and route to SMT queue, or permit frame-only purchase?	Establishes clinical guardrails and prevents lab remakes.
Farhan (Ops)	What is the store SOP if network reconnection exceeds 15 minutes post-terminal payment?	Guides offline reconciliation and customer receipt handling.

2. Scope Matrix: Building vs. Deferring

In Scope for v1 (Must Requirements)

1. **Customer & Rx Lookup:** Phone search via Shopify; latest valid Rx from SMT (`/smt/prescriptions`). Cohorts (`priority`, campaigns) from Shopify customer tags.
2. **Cart & Lens Matrix Validation:** Real-time validation of sphere, cylinder, and ADD against `rx_limits`, blocking invalid picks with clear staff prompts.
3. **Automated Offer Engine:** Date-effective rules from `offers.json` (stackability, priority, tags, pair bundles).
4. **Date-Effective GST Engine (by HSN):** Backward tax calculation from MRP to the paisa across 5 categories in `tax_rates.json`:
   - **HSN 90031900** (Frames): 18% (1800 bps) | **HSN 90015000** (Lenses): 5% (500 bps from 2025-09-22; 12% prior)
   - **HSN 90049000** (Sunglasses): 18% (1800 bps) | **HSN 90041000** (Corrective spectacles unit): 5% (500 bps)
   - **HSN 998599** (Services: eye test / fitting): 18% (1800 bps)
5. **Durable Order Creation:** Idempotent terminal charge followed by Shopify order creation carrying line attributes (`pair_ref`, `rx_id`, `for_customer_id`, HSN) and local sequential invoice numbers in `note_attributes`.
6. **Full Refund & Targeted Restock:** Line-level tax reversal, credit notes, frame returned to billing store stock, and custom lens scrapped without restock.
7. **Day-End Store Reconciliation Report:** Daily report for store and date from Shopify orders (bill count, gross, taxable, CGST/SGST, discounts, refunds, payment splits).

Explicitly Deferred to v1.1 (Should Requirements & Non-POS Tasks)

- **Sales Channel Capture & Reporting (Should):** Tagging orders (`walk_in`, `whatsapp_booked`, `online_pickup`) and store channel reports.
- **Cash Till Sessions & Drawer Tracking (Should):** Opening float, petty cash paid-outs (tea, courier), paid-ins, and drawer variance.
- **Full Remake Workflow (Should/Stretch):** Commercial liability allocation (lab vs. store vs. customer). Lens replacements bill referencing `parent_invoice_no`.
- **Automated 10 AM Daily Email Digest:** Handled by a backend cron querying Shopify Admin API, outside the till client.
- **Queue Tool & Facilities Maintenance:** SMT questionnaire updates and Indiranagar footfall camera hardware repairs.

3. Contradictions, Risks & Proposed Solutions

A. The <10-Second Target & Measurement

We cannot promise a sub-10s checkout without measurement. Estimated latency breakdown:
`Total Time ≈ Staff Input + Local Calc (<10ms) + Terminal Roundtrip (~2–5s unverified) + Shopify API (~1–2s unverified)`
Telemetry will track p50/p95 times. Speed will come from in-memory processing and catalog caching, never by skipping audit controls.

B. Manager Discounts & Non-Repudiable Attribution

The 10% cap requires formal Finance/Retail agreement. Overrides require authentication by a manager with `can_override_discount: true` in `staff.json`, recording approver `staff_id`, discount amount, and reason code (`WALKOUT_RESCUE`, `DAMAGE_ALLOWANCE`, `PRICE_MATCH`). No shared PINs.

C. Family Billing with Single Customer Constraint

One primary Shopify customer per order. Line items carry `pair_ref`, `rx_id`, and `for_customer_id` properties. Cross-account loyalty is deferred to v1.1.

D. Invoice Lifecycle & Immutability

Sequential invoice numbers are generated atomically from local SQLite with strict states:

- `POSTED`: Captured on terminal and confirmed in Shopify.
- `PENDING_RECONCILIATION`: Terminal funds captured; awaiting Shopify sync. **Never treated as cancelled; number is never reused.**
- `CANCELLED`: Aborted pre-payment; retained for gapless tax audit sequences.

E. Network Resilience & Safe Distributed Reconciliation

Shopify's REST API `/orders.json` lacks idempotency keys; blind retries risk duplicate orders and double-decremented stock. Furthermore, **an immediate lookup returning no order does not prove creation failed**, as downstream indexing delays occur. **Workflow:** (1) Persist local intent with `invoice_no`. (2) Idempotent terminal charge (`order_ref = invoice_no`). (3) Store `charge_id` locally; mark invoice `PENDING_RECONCILIATION`. (4) Submit order to Shopify. If response is ambiguous (timeout, network drop), keep invoice `PENDING_RECONCILIATION` and reconcile by listing recent orders at that `location_id` and inspecting `note_attributes` locally (as Shopify lacks direct `note_attributes` filtering). Never blindly replay; retry only if failure is conclusively established. (5) Transition to `POSTED` upon confirmed sync.

F. Prescription Source of Truth & Axis Validation

Shopify customer `tags` supply priority/campaign markers; SMT supplies prescriptions. SMT entry forms must enforce the 5° grid; POS will display an informational advisory for off-grid external prescriptions (e.g. 172°) without blocking checkout.

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4. Production Dependencies & Launch Checklist

1. **Shopify Admin API Credentials:** Custom App token with scopes: `read_products`, `read_customers`, `read_inventory`, `write_inventory`, `read_orders`, and `write_orders`, plus verified rate limits.
2. **Payment Terminal Integration:** SDK/API specs, idempotent retry docs, and payment terminal refund API and reconciliation documentation.
3. **Staff Identity & SMT Services:** Production SMT endpoints for `/smt/prescriptions` and `/smt/staff`, including staff authentication and permission verification.
4. **Finance & Tax Compliance:** Approved FY naming convention, credit note prefix, sign-off on date-effective HSN rates, and confirmation of GST e-invoicing applicability, IRN generation, B2B/B2C reporting, and credit-note filing rules.
5. **Hardware Specifications:** Driver specifications for ESC/POS thermal receipt printers and USB/Bluetooth barcode scanners.

The engineering team is ready to begin Phase 1 upon approval.